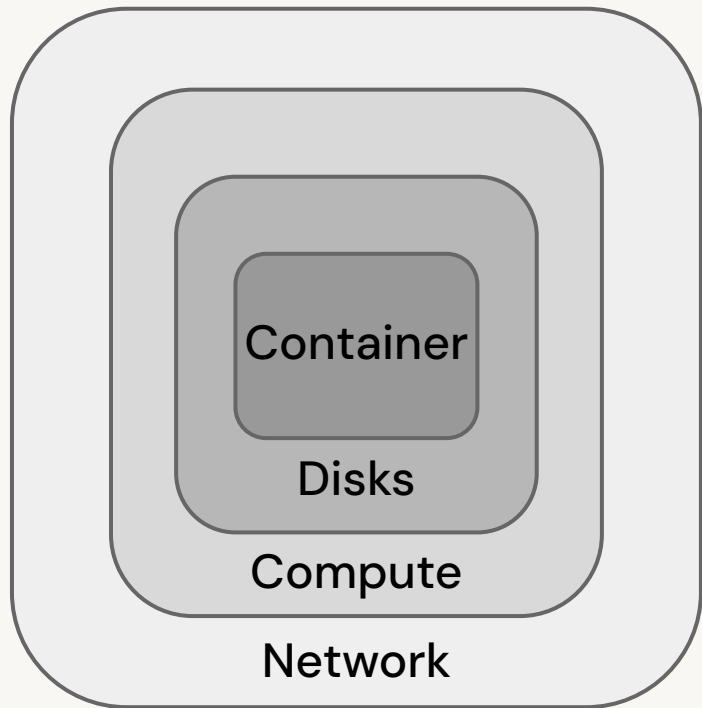


Multiple Layers of Isolation

Customer workloads are protected by multiple layers of isolation



Container

- Workloads run in an unprivileged container

Disks

- All local and attached disks are dedicated to a customer & wiped after use
- All disks are temporary & encrypted at rest

Compute

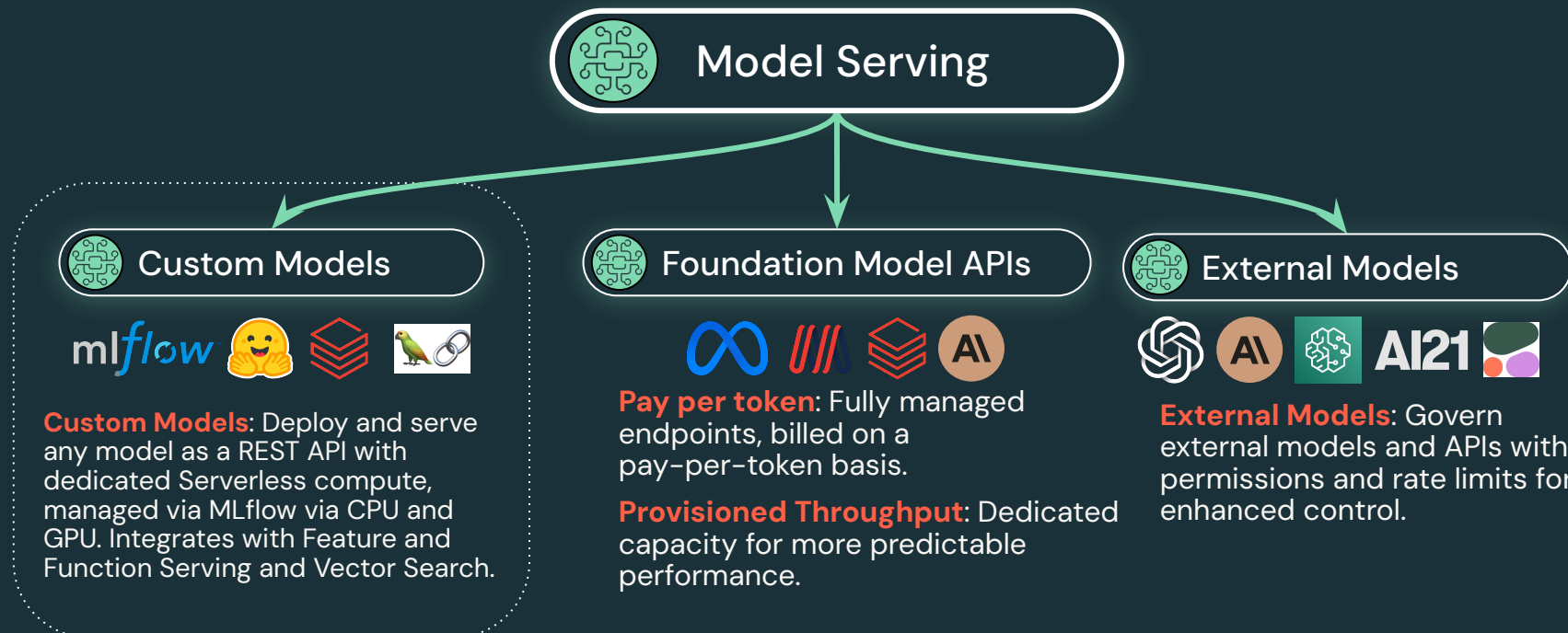
- Dedicated to a customer & wiped after use
- No privileges or credentials for other systems

Network

- Each workload / cluster is on an isolated logical network
- No public IP addresses
- No ingress allowed from other workloads / clusters

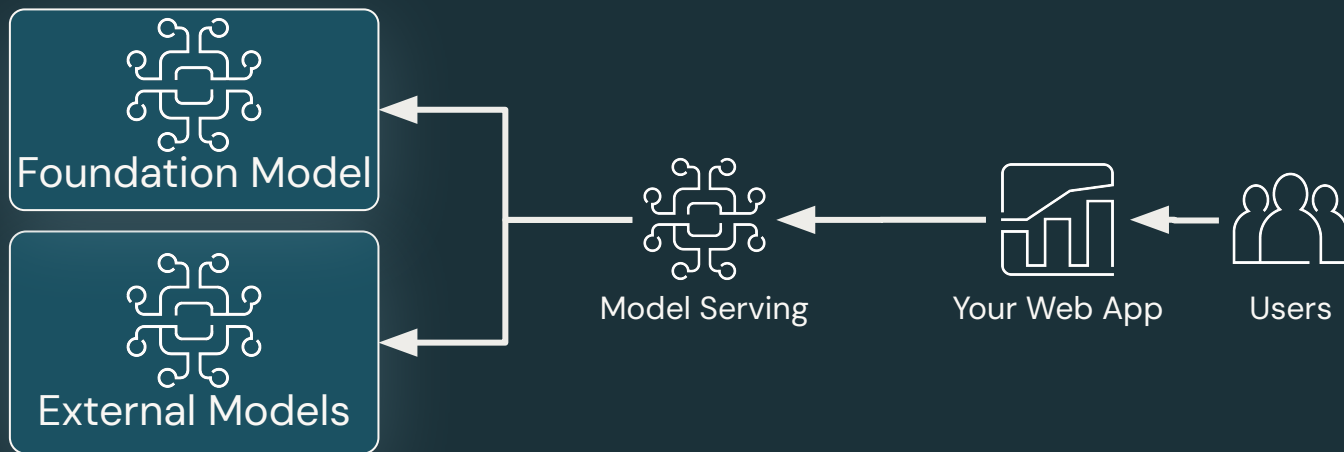
Understanding Model Serving

Databricks model serving allows for all types of models to be served and managed.

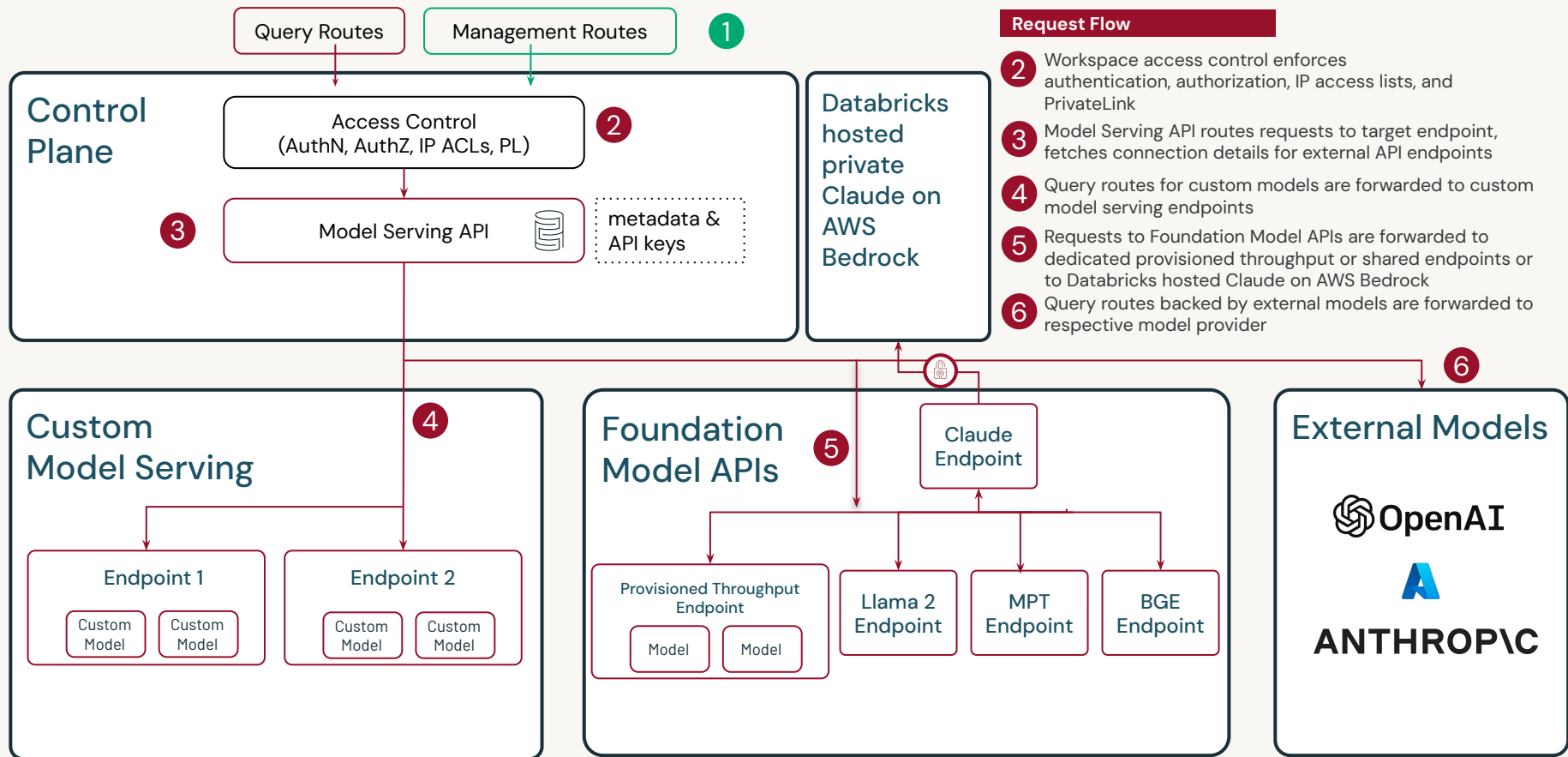


Foundation & External Model APIs

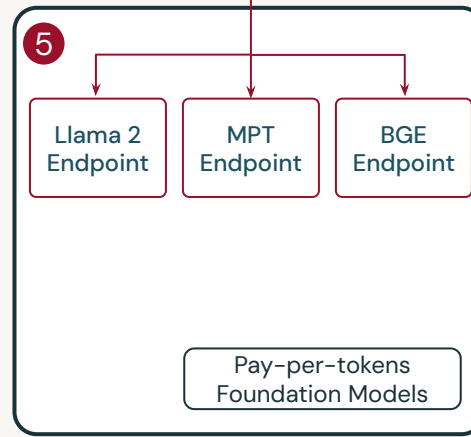
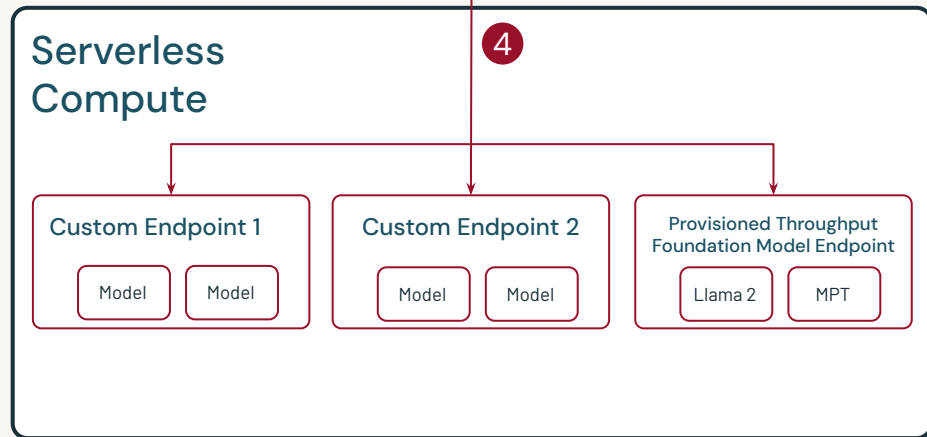
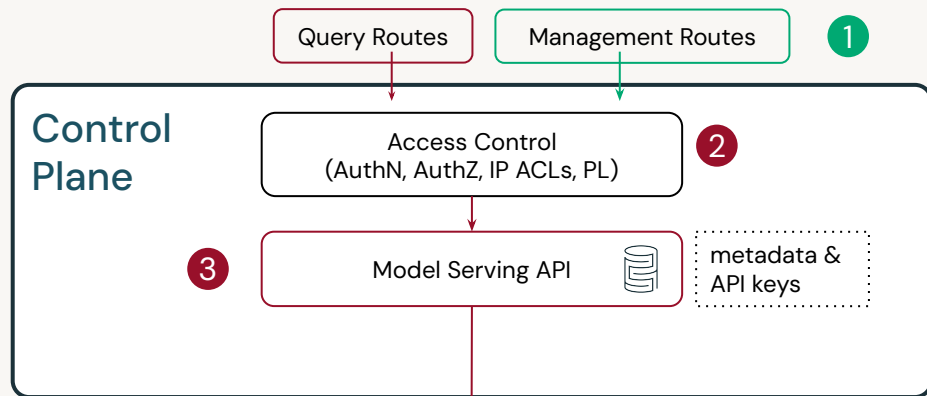
- Foundation Models provide Databricks–curated open source models
 - Hosted either pay-per-token or with provisioned throughput.
 - **Stateless, Trusted Code, Endpoints do not persist data or context.**
- External Models provide a layer of governance over commercial models hosted by companies like OpenAI or Anthropic.



Serving LLMs



Security: Overall Architecture



Management Flow

- 1 Manage endpoint configs, rate limits, API keys for remote models, ACLs, and compute for custom or provisioned models

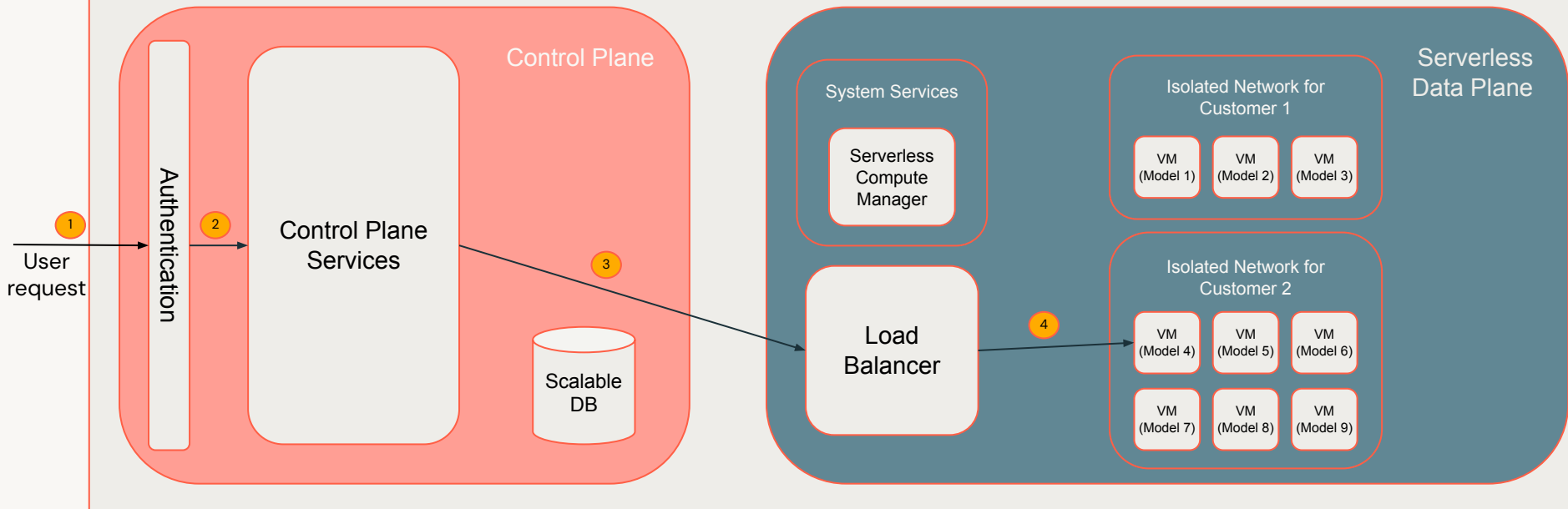
Request Flow

- 2 Workspace access control enforces authentication, authorization, IP access lists, and PrivateLink
- 3 Model Serving API routes requests to target endpoint, fetches key and URL for remote API endpoints
- 4 Query routes for custom models or provisioned Foundation Models forwarded to Serverless compute
- 5 Query routes backed by Foundation Model endpoint is forwarded to Databricks-managed models
- 6 Query routes backed by remote models is forwarded to respective external model provider

Security: Deep Dive

* only applies to Custom and Provisioned Throughput Models

Customer's Databricks Account



Any network communication between two services (1,2,3 and 4) is encrypted.

A VM is never reused and is destroyed after use.



